9.6 Matter Is Conserved in Mixtures

4th Grade Science · Video: Conservation of Matter Video for Kids, Mini-Clip (Generation Genius)

Name: _____ Date: _____

Watch the video and fill in the blanks as you go.

The Law of Conservation of Matter

The law of conservation of matter says that the amount of matter stays the _____, even when it changes form.

Matter never just _____.

The Experiment

First, they weighed out _____ grams of sugar.

Next, they weighed out _____ grams of water.

The scales were set so they don't weigh the _____. We want just the weight of the sugar and the water.

Then they _____ the sugar into the water.

Where Did the Sugar Go?

It looks like the sugar disappeared, but it just _____.

What evidence do we have that the sugar is still there? We can _____ it.

If the mixture weighs 150 grams, the sugar is gone. If it weighs _____ grams, the sugar is still there.

The mixture weighed 175 grams. That is the weight of all the water _____ the sugar.

Even though we can't see it, the sugar is still in there. It didn't _____.

Do the Math

25 grams of sugar + 150 grams of water = _____ grams of sugar water.

Mass of the mixture = mass of the first part + mass of the _____ part.

Vocabulary

Word	Meaning
matter	anything that has mass and takes up space
mixture	two or more kinds of matter put together
dissolve	to mix into a liquid so completely that you can't see it anymore
mass	how much matter something has; measured in grams
conservation of matter	the amount of matter stays the same, even when it changes form or is mixed

Key Takeaways

• When you make a mixture, the total mass is the _____ as the masses of the parts added together.

• Dissolving can make matter hard to see, but the matter is still there.